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Binding mechanisms of intrinsically disordered proteins: Insights from experimental studies and structural predictions

Thibault Orand and Malene Ringkjøbing Jensen

Advances in the characterization of intrinsically disordered proteins (IDPs) have unveiled a remarkably complex and diverse interaction landscape, including coupled folding and binding, highly dynamic complexes, multivalent interactions, and even interactions between entirely disordered proteins. Here we review recent examples of IDP binding mechanisms elucidated by experimental techniques such as nuclear magnetic resonance spectroscopy, single-molecule Förster resonance energy transfer, and stopped-flow fluorescence. These techniques provide insights into the structural details of transition pathways and complex intermediates, and they capture the dynamics of IDPs within complexes. Furthermore, we discuss the growing role of artificial intelligence, exemplified by AlphaFold, in identifying interaction sites within IDPs and predicting their bound-state structures. Our review highlights the powerful complementarity between experimental methods and artificial intelligence-based approaches in advancing our understanding of the intricate interaction landscape of IDPs.

Addresses
Univ. Grenoble Alpes, CEA, CNRS, IBS, Grenoble, France

Corresponding author: Jensen, Malene Ringkjøbing (malene.jensen@ibs.fr)

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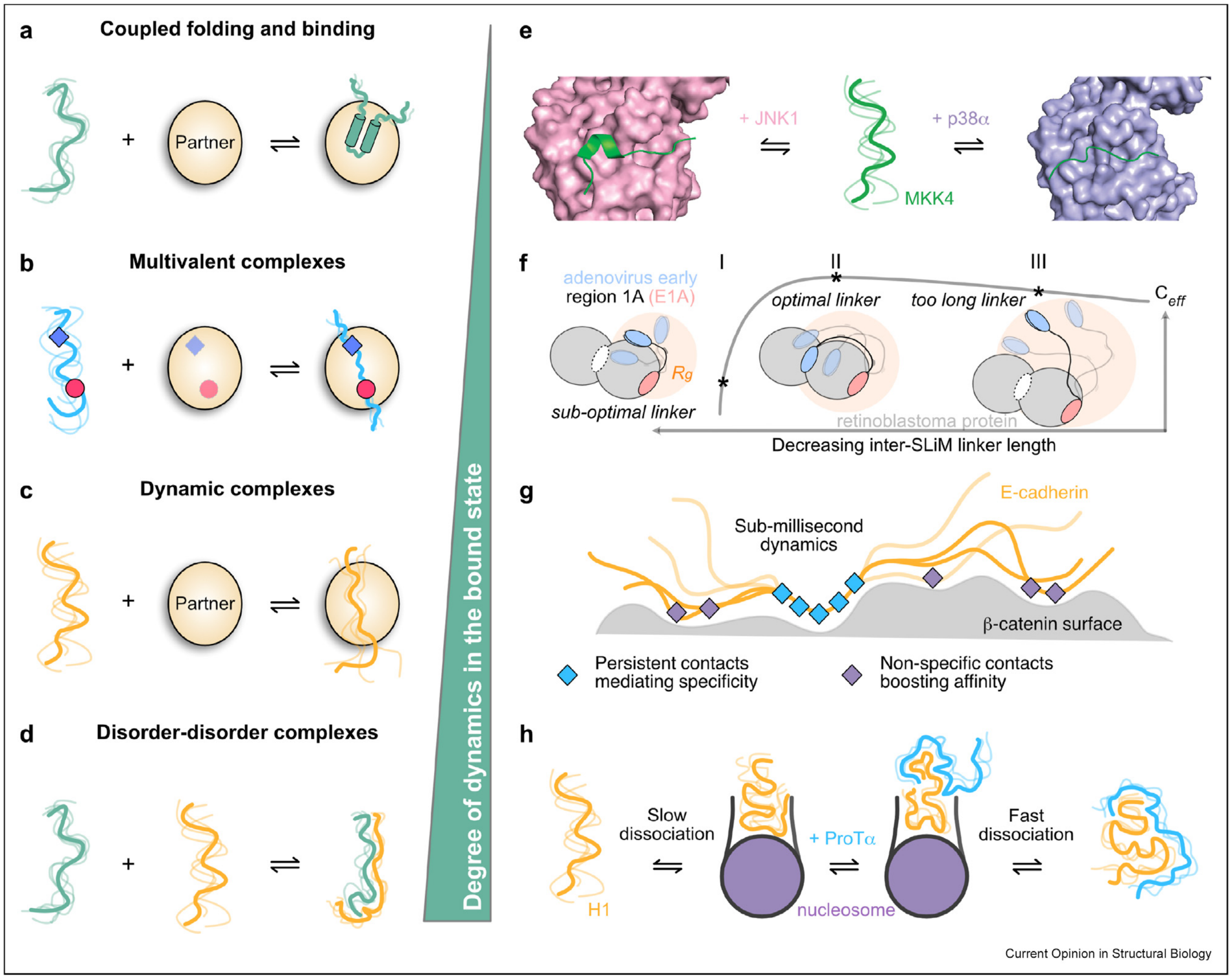
Introduction

Intrinsically disordered proteins (IDPs) or regions (IDRs) are highly abundant in the human proteome and serve as key regulatory components in many biological processes such as transcription, signal transduction, and cell cycle control [1,2]. Dysregulation of these essential cellular functions is linked to various diseases [3], underlining the importance of elucidating the structural and dynamic details of IDPs and their interactions with molecular partners.

It is now widely recognized that IDPs can employ a broad range of binding mechanisms, resulting in complexes with affinities that vary over several orders of magnitude [4]. IDPs typically make use of short linear motifs (SLiMs) for interacting with their folded binding partners [5], often leading to interactions involving coupled folding and binding [6]. This implies a transition from a conformationally heterogeneous free state to a well-defined, folded bound state with secondary or tertiary structure. Remarkably, IDPs can also retain conformational heterogeneity in the bound state *i.e.* they do not adopt a single, well-defined structure, but instead sample different conformations on the surface of their partners [7]. This can occur via a single SLiM or through multiple motifs that dynamically engage with the partner, as seen in multivalent complexes [8]. Finally, interactions between two disordered proteins have also been observed, however, instead of relying on specific linear motifs, these complexes are stabilized by rapidly exchanging, nonspecific electrostatic interactions between the protein chains [9,10]. Consequently, IDPs display a wide spectrum of dynamics in complex with their partners, ranging from fully folded conformations to entirely disordered states (Figure 1a–d).

Here, we review recent studies that illustrate this "dynamic spectrum" of IDP complexes highlighting the insights gained from a variety of experimental techniques. In particular, we focus on studies that elucidate the binding mechanisms of IDPs, focusing on the molecular details of their interaction trajectories, the characteristics of their transition states, and the structural dynamics of their final bound conformations. Furthermore, we highlight recent advances in predicting the bound-state structures of IDPs using state-of-the-art artificial intelligence (AI) tools such as AlphaFold, and we discuss how these tools can aid experimental design and deepen our understanding of IDP binding mechanisms.

Figure 1



Classification of IDP binding mechanisms **(a–d)** and schematic representations of experimentally characterized mechanisms belonging to each class **(e–h)**. The binding mechanisms can be divided into four main classes: **(a)** Coupled folding and binding where the IDP transitions from a conformationally heterogeneous, free state to a folded, bound state with secondary or tertiary structure. **(b)** Multivalent complexes where the IDP contains more than one SLiM for mediating interactions. The motifs may bind simultaneously to different binding pockets on the surface of the partner (as illustrated), or undergo exchange between different conformations with only one motif bound at any given time. **(c)** Dynamic complexes where the IDP remains highly dynamic in the bound state and samples multiple conformations on the surface of its partner. **(d)** Disorder-disorder complexes where two IDPs interact with each other forming a highly dynamic complex exchanging between multiple conformations. **(e)** Example of coupled folding and binding: The disordered regulatory domain of MKK4 folds into two distinct conformations upon interaction with the two homologous kinases, JNK1 and p38α [19]. **(f)** Example of a multivalent complex: The adenovirus early region (E1A) protein interacts through two binding motifs with the retinoblastoma protein. The length and sequence composition of the hypervariable linker connecting the two motifs are optimized through evolution to ensure maximum binding affinity [27]. **(g)** Example of a dynamic complex: E-cadherin samples multiple conformations on the surface of β-catenin on the sub-millisecond time scale. Persistent contacts of β-catenin with the central region of E-cadherin mediate specificity, while non-specific contacts with the termini of E-cadherin boost affinity [32]. **(h)** Example of a disorder-disorder complex: The linker histone H1 interacts with the nucleosome retaining sub-microsecond dynamics in the complex. Interaction of H1 with prothymosin α (ProTα) facilitates faster dissociation of H1 from the nucleosome through a "competitive substitution" mechanism [33].

NMR spectroscopy reveals atomic-level details of IDP interactions

Various experimental techniques have been employed to elucidate the binding mechanisms of IDPs — each offering a unique piece of the puzzle. Among these, nuclear magnetic resonance (NMR) spectroscopy stands out due to its ability to provide atomic resolution structural information and dynamics across various time scales [11–16]. Thus, NMR is a powerful technique to follow structural transitions in IDPs upon binding, to reveal interactions relying on multivalency and to detect the presence of low-populated, transient states or encounter complexes that may be crucial for IDP binding.

For example, Sekhar and co-workers used NMR spectroscopy to investigate the binding mechanism of the disordered DNA-binding domain of the bacterial regulator CytR with cognate DNA [17]. They reveal that CytR is predominantly disordered, but contains transiently populated helical elements that exchange into a higher-energy, excited state forming a three-helix bundle. Using NMR chemical exchange saturation transfer experiments, they identify the DNA-binding competent species as this folded, excited state of CytR. Their work represents a rare example of an IDP interaction that mainly relies on a folding-before-binding conformational selection mechanism.

Clore and colleagues employed a combination of solid-state NMR and electron paramagnetic resonance (EPR)-based double electron-electron resonance (DEER) to study the interaction of the disordered peptide M13 from myosin light-chain kinase with calmodulin [18]. Utilizing a rapid (millisecond time scale) mixing and freezing technique to trap different conformational states, they reveal that M13 can initially form two distinct encounter complexes while remaining disordered: one with the C-terminal domain of calmodulin (A state) which transitions directly to the final, fully-folded, bound state (C state), and another with the N-terminal domain of calmodulin (A* state), which requires a transition through a partially ordered intermediate (B state) before reaching the final bound state (C state). The study shows that calmodulin remains in an extended state in the encounter states and undergoes a conformational change to a compact state in the C state. The work illustrates the power of combining solid-state NMR and EPR for studying interaction mechanisms in a time-resolved manner.

IDPs can engage in promiscuous interactions with different binding partners. One recent example is the interaction of the disordered regulatory domain of the kinase MKK4, whose docking site motif (D-motif) can recruit two different mitogen-activated protein kinases (MAPKs), p38 and JNK [19]. NMR exchange techniques reveal that the D-motif of MKK4 adopts an extended structure upon binding to p38 [20], whereas it folds into a short α-helix upon binding to JNK1 [19] (Figure 1e). Despite the binding surfaces of both MAPKs being largely conserved, subtle sequence differences lead to variations in the architecture of the D-motif binding pocket, which, in turn, cause the differential folding of the motif [21]. This demonstrates that the different bound-state conformations of the D-motif are encoded not only within its sequence, but are also directed by the structural features of the binding pocket on the interaction partner, a concept known as "templated folding" [22,23].

IDPs can use a combination of multiple SLiMs for interacting with their partners. This may have the advantage of reinforcing affinity and selectivity of their interactions [8,24]. In this context, Wright and co-workers studied the competition mechanism between the disordered proteins HIF-1α and CITED2 for binding to the TAZ1 domain of the transcriptional coactivators CBP and p300 [25]. HIF-1α is a transcription factor involved in the hypoxic response, while CITED2 acts as its negative regulator. CITED2 contains three SLiMs that bind to TAZ1: an N-terminal αA region (forming a stable helix in the bound state), an LPEL motif, and a φC motif (a highly conserved hydrophobic and acidic region). When CITED2 is added to TAZ1:HIF-1α complexes, the αA region of CITED2 interacts in fast exchange with the αA-binding site on HIF-1α, forming an intermediate, ternary complex. Subsequently, the LPEL and φC motifs of CITED2 cooperatively displace HIF-1α. Once dissociated, HIF-1α cannot rebind and displace CITED2 from TAZ1, indicating a multivalency-driven unidirectional competition effect involving a two-step binding mechanism of CITED2 to HIF-1α.

Another IDP that relies on multivalency is ANP32A, a transcription regulator crucial for the adaptation of avian influenza virus polymerase to human hosts. ANP32A colocalizes the viral nucleoprotein NP and the 627-NLS domain of the PB2 subunit of the polymerase. Blackledge and colleagues used NMR to show that avian ANP32A binds NP at a distinct site from 627-NLS [26]. In contrast, human ANP32A, which lacks a 33-amino acid insertion present in the avian variant, exhibits overlapping binding sites for NP and 627-NLS. This results in a highly dynamic ternary complex that relies on multivalent electrostatic interactions across a 50-amino acid stretch of the IDR of ANP32A. The differing modes of interaction between the avian and human variants potentially explain why adaptive mutations in the 627-NLS domain, that enhance its basic interaction surface, are necessary for human ANP32A to efficiently colocalize NP with the viral polymerase.

Despite sequence variations, IDRs can retain functional traits throughout evolution. One example is the adenovirus early region 1A (E1A) protein [27]. E1A competes with host factors to bind the retinoblastoma (Rb) protein using two SLiMs separated by a hypervariable, flexible linker. The simultaneous interaction of both motifs with Rb results in picomolar binding affinity, which is more than two orders of magnitude stronger than the affinity observed when each motif binds separately. Using a combination of NMR, small angle X-ray scattering (SAXS), biophysical techniques, and molecular dynamics, Chemes and colleagues demonstrated that this synergistic increase in binding affinity arises from an increased local effective concentration, where binding of one motif enhances the local concentration of the other motif on the surface of Rb (Figure 1f). Intriguingly, sequence analysis of mastadenoviruses reveals variability in the length and composition of the linker region questioning the conservation of this tethering effect. The authors show that covariations in the linker sequence length and composition occur to preserve the ensemble dimensions of the linker, thereby maintaining an optimal tethering effect — a mechanism they term "conformational buffering". The study provides a striking example of how hypervariable IDRs can be under evolutionary pressure to conserve their sequence-ensemble relationship for optimal functional interaction.

Taken together, the above examples underline the power of NMR spectroscopy in characterizing the versatile interaction landscape of IDPs, with a major advantage being its ability to provide atomic-level resolution.

Single-molecule techniques elucidate the dynamics of IDP complexes

In addition to NMR spectroscopy, single-molecule techniques are valuable for elucidating the binding mechanisms of IDPs [28—30]. In particular, these techniques can enable the detection of stochastic events within heterogeneous transition pathways during coupled folding and binding, provide a mapping of IDP dynamics on the surface of their interaction partners and reveal novel interaction mechanisms of IDPs that are otherwise challenging to observe using ensemble-averaged techniques.

Chung and Kim studied the disordered transactivation domain (TAD) of the tumor suppressor p53 that folds upon binding to the nuclear coactivator binding domain (NCBD) of the CREB binding protein [31]. They used fast three-color single-molecule Förster resonance energy transfer (smFRET) to identify two main transition pathways, TP1 and TP2, representing 50—60 % and 40—50 % of transitions, respectively, and to measure the associated transition path times. TP1 has longer lifetimes (300—800 μs) and is stabilized by strong electrostatic interactions between the C-terminal part of TAD and NCBD, while TP2 has shorter lifetimes (16—24 μs) and is characterized by weaker electrostatic contacts between the central region of the TAD and NCBD. Interestingly, these electrostatic interactions are not present in the final bound state, suggesting that the binding proceeds via early nonnative contacts followed by late formation of native contacts. Their work provides an excellent demonstration of how IDPs can exploit heterogeneous transition paths during coupled folding and binding.

IDPs can remain dynamic when complexed with their partners, however, the time scales and spatial amplitudes of such motions have remained largely unexplored. In their recent work, Hofmann and colleagues studied the complex between the disordered tail of E-cadherin and β-catenin using a combination of smFRET and molecular simulations [32]. They demonstrate that E-cadherin diffuses on the surface of β-catenin on the sub-millisecond timescale and visualize the conformational space adopted by E-cadherin within the complex. This diffusion mechanism is facilitated by persistent interactions from the core region of E-cadherin, which mediate specificity, while weak contacts from the surrounding disordered regions provide additional binding affinity (Figure 1g). The study highlights the utility of smFRET in revealing both the time scales and amplitudes of the conformational dynamics of IDPs in complex with their partners.

Another example of an IDP remaining dynamic in its bound state is the highly positively charged and largely disordered linker histone H1 that interacts with nucleosomes. Schuler and colleagues investigated how the interaction kinetics of H1 are modulated by the presence of another disordered protein, prothymosin α (ProTα), which is highly negatively charged [33]. Using a coarse-grained model of the H1-nucleosome complex, aligned with experimental smFRET data, they find that the disordered regions of H1 retain sub-microsecond dynamics within the complex. Further simulations, incorporating ProTα, reveal that H1 recruits ProTα into a ternary complex. ProTα competes with DNA for interaction with H1, leading to an opening of the DNA arms and promoting dissociation of H1 from the nucleosome (Figure 1h). This study highlights a fascinating "competitive substitution" mechanism driven by electrostatic interactions between two highly charged and disordered proteins.

IDPs interact not only with other proteins but also frequently form complexes with nucleic acids. A recent example is the pioneer transcription factor Sox2, which contains a structured DNA-binding domain (DBD) followed by a C-terminal IDR that includes two activation domains (AD1 and AD2). Heidarsson and co-workers used a combination of NMR and smFRET to study how DNA binding influences the conformational ensemble of Sox2 [34]. They find that the IDR's ensemble dimensions are shaped by electrostatic interactions with the positively charged DBD. Interestingly, regions around AD1 and AD2 are identified as key sites of interaction with the DBD, suggesting that these long-range electrostatic interactions help organize Sox2's functional regions. DNA binding significantly expands the structural ensemble of the IDR (increasing the root-mean-square distance by 20 %) leading to a differential exposure of the two activation domains. A similar expansion of the ensemble occurs upon nucleosome binding, highlighting potential relevance in a chromatin context. Overall, the study suggests that dynamic electrostatic intramolecular interactions, and their rearrangement upon DNA binding, could represent a common mechanism in transcription factors essential for fine-tuning their structural ensembles to regulate function.

New single molecule techniques have recently been developed for studying IDPs and their interaction mechanisms. Guo and colleagues have developed an electrical biosensor using silicon nanowire field-effect transistors for in-situ, label-free, real-time detection of single-molecule events [35]. This biosensor was employed to study structural transitions in the partially disordered protein c-Myc and its interaction with its partner Max. The authors reveal that c-Myc exchanges between disordered and partially folded states, and that its binding to Max is preceded by the formation of an encounter state in c-Myc. This encounter state facilitates complex formation via an induced-fit mechanism. This sensitive and time-resolved technique holds great promise for application to other disordered protein systems.

In summary, the aforementioned studies highlight the value of single-molecule techniques for quantifying transition state lifetimes and for determining the amplitude and time scales of the dynamics of IDPs when interacting with their binding partners.

Stopped-flow fluorescence reveals structural properties of transition states

Kinetic studies, notably using stopped-flow fluorescence, have significantly enhanced our understanding of coupled folding and binding mechanisms of IDPs [36–39]. These methods are particularly suitable for distinguishing between induced-fit and conformational-selection contributions to binding [40,41], and when combined with protein engineering, they offer valuable insight into the structural properties of transition states.

Jemth and colleagues explored the evolution of the binding mechanism of the CREBBP-interacting domain (CID) from NCOA3 with the nuclear coactivator binding domain (NCBD) from the CREB-binding protein [42]. They compared the transition state properties of the high-affinity NCBD-CID complex in *Homo sapiens* with those of a reconstructed, low-affinity Cambrian-like complex using stopped-flow fluorescence, Φ-value analysis, and molecular dynamics simulations. The modern complex exhibits stronger affinity and greater ordering but fewer native contacts in the transition state compared to the Cambrian-like complex. This suggests that, although the binding mechanism has been conserved, the transition state is more heterogeneous and less native-like in the modern complex, offering novel insights into the evolution of binding mechanisms.

Using similar techniques, the interaction between the disordered region of the amyloid precursor protein and the phosphotyrosine-binding domain of Mint2 was investigated [43]. The study reveals that the rate-limiting transition state in the binding process involves both native and nonnative hydrophobic interactions as well as the formation of hydrogen bonds. These interactions form cooperatively during the coupled folding and binding process, with approximately one-third of native contacts established at the transition state. This complex stands out from previously studied IDP interactions due to its unusually strong reliance on hydrophobic interactions during the crossing of the transition state barrier.

Stopped-flow kinetics were also employed to study the interaction between the intrinsically disordered protein TCF7L2 and β-catenin, a critical interaction for Wnt signaling [44]. The experiments revealed that the complex association kinetics are monophasic, while the dissociation kinetics are biphasic. Alanine scanning indicates that the N-terminal and C-terminal regions of TCF7L2 are responsible for high-affinity binding to β-catenin, while the intermediate region remains flexible within the complex. The biphasic dissociation suggests a “Velcro-like” mechanism, where TCF7L2 sequentially unbinds from β-catenin by first dissociating either the N-terminal or C-terminal region. This study demonstrates how an IDP can utilize different mechanisms for association and dissociation.

Overall, the mentioned examples demonstrate how stopped-flow fluorescence provides valuable insights into the kinetics of IDP interactions and the degree of native contacts within transition states – states that are challenging to characterize structurally using other experimental techniques.

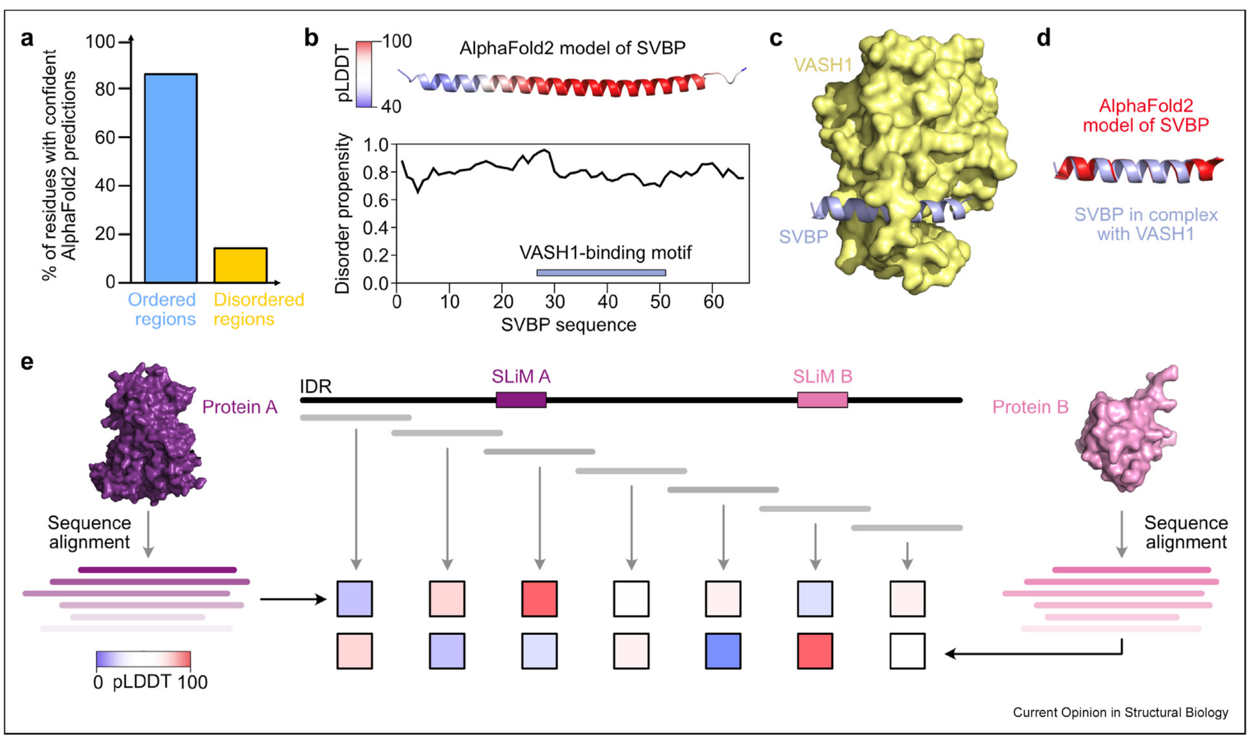
Structure prediction tools uncover the bound-state structures of IDPs

Recent advances in structure prediction of individual proteins or multi-protein complexes using AI have revolutionized structural biology [45–48]. However, IDPs and their complexes remain challenging targets for structure prediction. First of all, IDPs often yield low-quality multiple sequence alignments (MSAs) due to their rapidly evolving sequences [49], which consequently impact the accuracy of structural predictions. Secondly, IDPs are underrepresented in the PDB database of experimentally determined structures, limiting the availability of suitable structural templates. Despite these challenges, recent studies have shown that AI approaches, such as AlphaFold, can still contribute to our understanding of the conformational properties of IDPs and their complexes, as described below.

The AlphaFold Protein Structure Database provides structure predictions of proteins as single, static conformations. It is now widely recognized that protein regions with low confidence scores (predicted Local Distance Difference Test (pLDDT) < 50) correspond to segments of high conformational heterogeneity [50], making AlphaFold an efficient disorder predictor [51]. However, rather than viewing IDRs as a single conformation, they should be represented by an ensemble of structures. These ensembles can be derived from experimental data, such as NMR [52–57], SAXS [58,59], EPR [60,61], smFRET [62], or from combinations of these techniques [63–66]. Computational methods can also be used to generate these ensembles, as demonstrated by Lindorff-Larsen and colleagues who simulated nearly all IDRs in the human proteome and derived interesting relationships between hydrophobicity, charge patterning, ensemble compaction, subcellular localization and biological function [67]. Finally, we note that AlphaFold, as a stand-alone application, may generate multiple conformations of an IDP. While this conformational variability could reflect the inherent structural heterogeneity of IDPs, the resulting conformations rarely capture the true ensemble dimensions of IDPs.

Despite the recognition of conformational heterogeneity within IDRs, recent studies have shown that the single conformations available in the AlphaFold Database contain information about conditionally folded regions [68,69]. Forman-Kay and colleagues found that AlphaFold assigns high confidence scores (pLDDT >70 %) to about 15 % of human IDRs, and that the predicted structures of many (approximately 60 %) of these regions correspond to the bound-state conformations of IDPs known to fold upon binding [69] (Figure 2a–d). Interestingly, further proteomic analysis revealed that disease-related mutations are five times more prevalent in these conditionally folded regions, suggesting a direct link to function. These findings demonstrate the potential of systematically analyzing IDR sequences and their predicted structures to better understand the structural transitions associated with coupled folding and binding, and to identify novel binding sites for interaction partners.

Figure 2



Leveraging AlphaFold to identify conditionally folded regions in IDPs and to predict their complex structures. **(a)** A systematic analysis of the AlphaFold Protein Structure Database reveals that AlphaFold2 assigns high-confidence structures (pLDDT >70) to nearly 15 % of human IDRs. Many of these IDRs conditionally fold upon interaction with partner proteins, and AlphaFold2 predominantly predicts the structures corresponding to these folded states [69]. **(b)** As an example, the small vasohibin binding protein (SVBP) is predicted to be intrinsically disordered along its entire sequence (using flDPnn as a disorder predictor [76]), but AlphaFold2 predicts a folded structure with one major helical element. **(c)** Crystal structure of SVBP (purple) in complex with vasohibin 1 (VASH1, yellow) (PDB 6OCG [77]). The structure was released in 2019 and therefore not included in the AlphaFold2 training set. **(d)** The AlphaFold2-predicted structure of SVBP closely matches the conformation of SVBP in complex with VASH1. **(e)** Proposed “sliding window” approach for improving the accuracy of structure predictions of IDP complexes using AlphaFold2. Individual predictions are performed on smaller, overlapping IDP segments (gray) with a delimited sequence of the folded partner obtained on the basis of sequence alignments. Each prediction yields a confidence score, which is used to rank the segments and to identify the most likely binding site within the IDP. Benchmarking against a dataset of 42 IDP complexes has demonstrated that using smaller segments results in more accurate structure predictions compared to full-length IDP sequences [73].

One of the most powerful features of AlphaFold is its ability to accurately predict the structures of protein-protein complexes [70]. Recent studies have evaluated AlphaFold's performance in predicting the structures of complexes between IDPs and their folded binding partners [71–75]. Guerois and colleagues compiled a set of 42 IDP complexes, nonredundant with the AlphaFold training set, and systematically explored various MSA strategies to assess their impact on the accuracy of the predicted structures [73]. They found that restricting the sequences to the precise binding regions of both the IDP and its partner significantly enhanced accuracy. Using full-length sequences, the success rate was 43 %, but this increased to 79 % when both sequences were delimited and their respective MSAs were included in the prediction. Similar observations were made by Gsponer and colleagues who compiled a more extensive testing set characterized by larger IDP interaction interfaces [75]. Delimiting sequences can, however, be difficult in an experimental scenario where the binding site on the IDP is unknown. To address this, a "sliding window" approach was proposed (Figure 2e), where structure predictions are made and ranked using smaller segments of the IDP, typically around 100 amino acids in length [73,74]. This method shows a high success rate, suggesting a broad applicability of this approach for predicting the bound structures of a wide range of SLiMs embedded in IDRs.

Conclusions

IDPs exploit a diverse array of binding mechanisms that we can broadly classify as coupled folding and binding, multivalent interactions, dynamic complex formation, or disorder-disorder interactions (Figure 1a–d). In this review, we have discussed recent examples of IDP binding mechanisms elucidated by various experimental techniques. It is clear that a simple classification is not straightforward and that many IDP complexes are formed via a combination of mechanisms that are not mutually exclusive. For example, an IDP can engage in coupled folding and binding through a SLiM, while the remainder of the chain interacts with the surface of the partner in a dynamic, nonspecific manner. Moreover, the interaction trajectories of IDPs may involve a heterogeneous ensemble of transient encounter complexes or intermediates characterized by varying degrees of native contacts that are not always observed in the final bound conformation. Integration of experimental techniques and the development of new methodologies hold great promise for further exploring the structural, kinetic and dynamic details of the discrete states populated along the interaction trajectory of IDPs.

There is growing interest in leveraging AI approaches for structural studies of IDPs and their complexes. In this review, we have discussed recent advances in the literature demonstrating that, despite the challenges posed by lower-quality MSAs for IDPs, AlphaFold remains a powerful tool for identifying protein interaction sites within IDPs and for revealing structural transitions upon binding by elucidating the molecular mechanisms that stabilize the final bound state. As such, AlphaFold is poised to continue playing a significant role in IDP research. However, it is also evident that AlphaFold currently lacks a "dynamic component," offering primarily a static view of IDPs and their complexes. Therefore, it is crucial for AlphaFold to be used in conjunction with experimental techniques to fully advance our understanding of this fascinating class of proteins.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data were used for the research described in the article.

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The binding of the intrinsically disordered regulatory domain of the kinase MKK4 to two homologous protein partners is studied. The authors show using NMR chemical exchange experiments that the same linear motif within MKK4 folds into two different conformations on the surface of the partners, while maintaining comparable binding affinities. This differential folding is shown to be consistent with measured thermodynamic profiles of the two interactions obtained from isothermal titration calorimetry. The study highlights the importance of the binding pocket architecture in dictating the final bound conformation of IDPs.

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